Pedagogical Transformation Using Large Language Models in a Cybersecurity Course

Abstract

Large Language Models (LLMs) are increasingly used in higher education, but their pedagogical role in fields like cybersecurity remains under-investigated. This research explores integrating LLMs into a university cybersecurity course using a designed pedagogical approach based on active learning, problem-based learning (PBL), and computational thinking (CT). Instead of viewing LLMs as definitive sources of knowledge, the framework sees them as cognitive tools that support reasoning, clarify ideas, and assist technical problem-solving while maintaining human judgment and verification. The study uses a qualitative, practice-based case study over three semesters. It features four activities focusing on understanding concepts, installing and configuring tools, automating procedures, and clarifying terminology, all incorporating LLM use in individual and group work. Data collection involved classroom observations, team reflections, and iterative improvements guided by action research. Results show that LLMs can provide valuable, customized support when students actively engage in refining, validating, and solving problems through iteration. LLMs are especially helpful for clarifying concepts and explaining procedures during moments of doubt or failure. Still, common issues like incomplete instructions, mismatched context, and occasional errors highlight the importance of verifying LLM outputs with trusted sources. Interestingly, these limitations often

act as teaching opportunities, encouraging critical thinking crucial in cybersecurity. Ultimately, this study offers empirical evidence of human-AI collaboration in education, demonstrating how LLMs can enrich active learning.